Project Plan: Decoding Mushroom Toxicity with Logistic Regression Analysis

1.Introduction

This project aims to analyze the Mushroom Dataset (UCI) using Logistic Regression with Gradient Descent. The dataset consists of 22 categorical features describing various characteristics of mushrooms and a target variable indicating whether the mushroom is edible or poisonous.

2. Dataset Overview

- **Source**: UCI Machine Learning Repository
- **Source link**: https://archive.ics.uci.edu/dataset/73/mushroom
- **Features**: 22 features (e.g., cap-shape, odor, gill color)
- **Instances**: 8124 instances with missing values
- **Target Variable**: Edibility (Edible vs. Poisonous)
- **Motivation**: The dataset is well-structured, lacks missing values, and is ideal for binary classification.

3. Algorithm Selection

- **Model**: Logistic Regression
- **Optimization Method**: Gradient Descent
- **Justification**:
    - Suitable for binary classification tasks.
    - Implementing gradient descent provides deeper insight into optimization processes.

4. Plan of Action

1. **Data Preprocessing**
    - Convert categorical features into numerical representations.
    - Normalize features if required.
    - Split dataset into training and testing sets.
2. **Model Implementation**
    - Implement logistic regression from scratch using gradient descent.
    - Experiment with different learning rates and convergence criteria.
    - Evaluate model performance using accuracy, precision.
3. **Insights & Conclusions**
    - Determine the most influential features for classification.
    - Analyze the effectiveness of logistic regression for this dataset.
    - Discuss potential misclassifications and their causes.

5. Expected Outcomes

- A well-performing logistic regression model for mushroom classification.
- Feature importance analysis to identify key attributes influencing edibility.
- Performance summary and potential areas for improvement.

6. Next Steps

- Implement and validate the model.
- Analyze results and refine the approach if necessary using Python.
- Summarize findings in the final report.